

Candidate No : _____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Bachelor of Computer Applications (BCA)
Semester-II University Examination September-2018
CA118 Advanced Programming

Date: 19/09/2018, Wednesday

Time: 10:00 am to 1:00 pm

Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. Figure to right indicates marks.
3. Make suitable assumption when necessary.

SECTION – 1

Q.1 Do as directed

[11]

1. What is nested structure?
2. What is the use of free() function?
3. What is the output of following code segment:
`int no=5000; int *ptr=&no; //&no=64000
printf("%u",ptr);`
4. What is the output of following code segment:
`int val[7]={67,89,90,201,206,900};
printf("%d", *(val+1));`
5. What is the output of following code segment:
`char c[14]= "cmpica is best";
printf("%s", (c+2));`
6. What is the output of following code segment:
`char c[10]= "BCA SEM2";
printf("%c", *c);`
7. Which operator is used to access member of a structure?
8. What is union?
9. What is enumerated Data Types?
10. Write any one difference between structure and union.
11. What is the structure pointer operator?

Q.2

- (A) What is static memory allocation? Differentiate between malloc() and calloc() functions. **[06]**
- (B) Explain the concept of Array of structures with example. **[06]**

OR

Q.2

- (A) Write a C program to read file line-wise. **[06]**
- (B) Explain following file opening modes with suitable example: **[06]**
(i) r+ (ii) w+ (iii) a+

Q.3

- (A) Explain about pointer and array with suitable example. **[06]**
- (B) Differentiate between binary file and text file. Explain following functions to read and write to the file: **[06]**
(i) fgets() (ii) fputs()

OR

Q.3

- (A) Explain about following file handling functions with suitable example: **[06]**
(i) fread() (ii) fwrite()
- (B) What is the use of fseek() function? Write a program to randomly access a file. **[06]**

SECTION – 2

Q.4 Answer the questions

[11]

1. What is the output of following code segment:

```
#define m(y) y*y
#include<stdio.h>
void main() {
    printf(“%d”,m(1-2)); }
```
2. How many fields are required in a singly linked list?
3. What is traversal in linked list?
4. Define circular linked list.
5. In stack, how many elements are there when TOP=0?
6. In queue, insertion is done at front end. Mention: True /False.
7. What is underflow in stack?
8. Write a line to include header file in a C program.
9. Which data structure works in FIFO manner?
10. What is the use of void pointer?
11. What is pre-processor?

Q.5

(A) Write algorithm for following operations on stack: [06]

- (i) PEEP (ii) CHANGE

(B) Convert following expressions from infix to postfix: [06]

- (i) $(a + b) * c + d$ (ii) $m / n + o - p$

OR

Q.5 Write following algorithms to insert a node in the singly linked list:

(A) (i) Insert node at front (ii) Insert node at the end [06]

(B) Differentiate between linked list and array. [06]

Q.6

(A) Give any two real life examples of queue. Write an algorithm to insert an element in the queue. [06]

(B) Why linked list is used? How linked list is represented in C? [06]

OR

Q.6

(A) A queue Q is given with F=1, R=3. Capacity (N) of queue is given as 5. The elements of queue are given as 11, 45, 50 front (F) is pointing towards 11 and rear (R) is pointing towards 50. Show the contents of queue when following operations take place: [06]

1. INSERT(Q, 120)
2. INSERT(Q, 100)
3. DELETE(Q)
4. DELETE(Q)
5. DELETE(Q)
6. INSERT(Q, 150)

(B) A stack S is given with TOP=1. Capacity (N) of stack is given as 5. The elements from the top of the stack are given as 80, 90, where top is pointing towards 80. Show the contents of stack when following operations take place: [06]

- (1) PUSH(S, 40)
- (2) POP(S)
- (3) POP(S)
- (4) PUSH(S, 120)
- (5) POP(S)
- (6) PUSH(S, 500)